

Sri Lanka Institute of Information Technology



Event Photography Videography Booking System

**Project Group : WD030
Y1S2 (2026 Jan - Jun)**

Object Oriented Programming | SE1020

B.Sc. (Hons) in Information Technology

Declaration

We hereby declare that this project report titled “Event Photography & Videography Booking System” is our original work developed for the SE1020 Object Oriented Programming module.

Acknowledgement

We would like to express our sincere gratitude to our lecturer and all individuals who supported us throughout the development of this project. Their guidance and encouragement helped us successfully complete this system.

Table of Contents

Introduction	04
Problem Statement	05
Objective	06
Scope of the System & Technologies Used	07
System Architecture	08-09
System Features	09-12
Database and File Handling	13
OOP Concepts Used	14-15
CRUD Operations	16
Class Diagram	17
Testing & GitHub Version Control	18
GitHub Repository & Commit History	19-20
Advantage of the System, Limitations & Future Enhancements	21
Conclusion	22

Introduction

The Photography & Videography Booking System is a web-based management application developed using the Spring Boot Framework and Java technologies. The purpose of this system is to simplify and automate the process of booking photography and videography services for different types of events such as weddings, birthdays, corporate events, parties, and engagement ceremonies.

Traditionally, photography bookings are managed manually through phone calls, physical records, and paperwork. This manual process can lead to scheduling conflicts, data loss, poor communication, and inefficient management of bookings and staff. The proposed system solves these problems by providing a centralized online platform where users can manage bookings efficiently.

The application follows the MVC (Model-View-Controller) architecture and applies major Object-Oriented Programming (OOP) concepts such as encapsulation, inheritance, abstraction, and polymorphism. The project also demonstrates proper file handling mechanisms using a custom file-based storage system instead of a relational database.

The system provides separate functionalities for customers and administrators. Customers can browse service packages, make bookings, select photographers or videographers, complete payments, and provide feedback. Administrators can manage events, service packages, bookings, staff details, and customer information through a dedicated admin panel.

This project was mainly developed as an educational project to demonstrate practical implementation of Java programming concepts, web technologies, software engineering principles, and backend development using Spring Boot.

Problem Statement

In many photography businesses, booking management is still performed manually using notebooks, spreadsheets, or phone records. These methods create several issues such as:

- Difficulty in tracking customer bookings
- Double-booking of photographers
- Poor management of service packages
- Lack of centralized customer records
- Difficulty in payment tracking
- Time-consuming paperwork
- Poor communication between customers and service providers

To overcome these limitations, the Event Photography & Videography Booking System was developed as an automated solution that digitizes and centralizes the entire booking process.

Objectives

The major objectives of the project are:

Primary Objectives

- To automate the manual booking system
- To reduce paperwork and human errors
- To provide an easy-to-use online booking platform
- To improve booking management efficiency
- To centralize all photography business operations

Educational Objectives

- To demonstrate implementation of OOP concepts
- To understand MVC architecture in Spring Boot
- To practice file handling in Java
- To develop knowledge of web application development
- To apply CRUD operations in a real-world system

Technical Objectives

- To implement secure authentication
- To manage persistent data using file handling
- To create modular and maintainable code
- To design reusable Java classes
- To implement layered application architecture

Scope of the System

The scope of the system includes:

- User Registration and Login
- Booking Management
- Event Management
- Service Package Management
- Staff Management
- Payment Tracking
- Feedback Management
- Admin Dashboard Operations
- File-based Data Persistence

The system is intended for small to medium-scale photography businesses that require a simple and efficient booking management solution.

Technologies Used

Technology	Purpose
Java 21	Backend Programming Language
Spring Boot 3.2.5	Backend Framework
HTML	Frontend Structure
CSS	Styling and Design
JSP	Dynamic Web Pages
Maven	Build Management
GitHub	Version Control
File Handling	Data Storage
MVC Architecture	System Design Pattern

System Architecture

The system follows the MVC (Model-View-Controller) architecture.

Model Layer

The model layer contains all entity classes such as:

- User
- Booking
- Staff
- Event
- Service Package
- Payment
- Feedback

These classes store the application data and business objects.

Controller Layer

The controller layer handles:

- HTTP Requests
- Form Submissions
- User Navigation
- Business Logic Processing

Examples:

- UserController
- BookingController
- PaymentController

View Layer

The view layer contains JSP pages responsible for:

- User Interfaces
- Forms
- Tables
- Dashboards
- Dynamic Content Rendering

System Features

User Management

The system provides secure user authentication and profile management.

Functions:

- User Registration
- User Login
- Password Reset
- Profile Update
- Account Deletion

Event Management

Administrators, Customers can manage different event types.

Supported Events:

- Wedding
- Birthday
- Engagement
- Corporate Event
- Baby Shower
- Party

Functions:

- Add Event
- Update Event
- Delete Event
- View Event Details

Service Package Management

The system supports multiple photography packages.

Example Packages:

- Silver Package
- Gold Package
- Platinum Package

Features:

- Package Price Management
- Package Description
- Editing Package Details

Booking System

The booking engine allows customers to reserve photography services online.

Booking Process:

1. Select Event
2. Choose Package
3. Choose Photographer/Videographer
4. Confirm Booking

Features:

- Real-Time Booking
- Booking History

- Booking Cancellation
- Booking Updates

Staff Management

Administrators can manage photographers and videographers.

Functions:

- Add Staff
- Update Staff Information
- Remove Staff
- Track Availability

Payment Management

The system tracks customer payments.

Features:

- Payment Recording
- Payment Status Tracking
- Booking Payment Association

Feedback System

Customers can provide reviews and ratings.

Functions:

- Submit Feedback
- View Reviews
- Manage Customer Ratings

8.8 Admin Dashboard

The administrator dashboard acts as the control center of the system.

Admin Functions:

- Manage Users
- Manage Events
- Manage Bookings
- Manage Packages
- Manage Staff
- View Reports
- Handle Payments

Database and File Handling

This system uses a custom file-based storage system.

File Storage Location

src/main/webapp/data/

Storage Format

Data is stored using:

- Text files
- Pipe-separated values (|)

Example: B1|U1|E1|P1|S1|50000

Reading Data

The system uses `BufferedReader` to read data line-by-line.

Example: `BufferedReader br = new BufferedReader(new FileReader(file));`

Writing Data

The system uses `FileWriter` and `PrintWriter`.

Example: `PrintWriter pw = new PrintWriter(new FileWriter(file, true));`

Advantages of File-Based Storage

- Simple implementation
- Easy to understand
- No external database configuration
- Lightweight storage solution

OOP Concepts Used

Encapsulation

All data members are declared as private and accessed through getter and setter methods.

Benefits:

- Data security
- Controlled access
- Better maintainability

Inheritance

Inheritance is used to reuse common structures and behaviors across classes.

Example:

- DAO classes follow similar structures
- Shared functionality can be reused

Benefits:

- Code reusability
- Reduced duplication
- Easier maintenance

Polymorphism

Method overriding is used extensively.

Benefits:

- Flexible implementation
- Customized behavior
- Better object representation

Abstraction

The application separates functionality into layers.

Layers:

- Models
- Controllers
- DAOs
- Views

Benefits:

- Reduced complexity
- Easier debugging
- Better scalability

CRUD Operations

Create Operations

- Register Users
- Add Events
- Add Packages
- Add Staffs
- Create Bookings
- Add Payments and Feedbacks

Read Operations

- View Profiles
- Display Events
- Display Pageages
- View Staffes
- View Booking History
- View Payment and Feedbacks

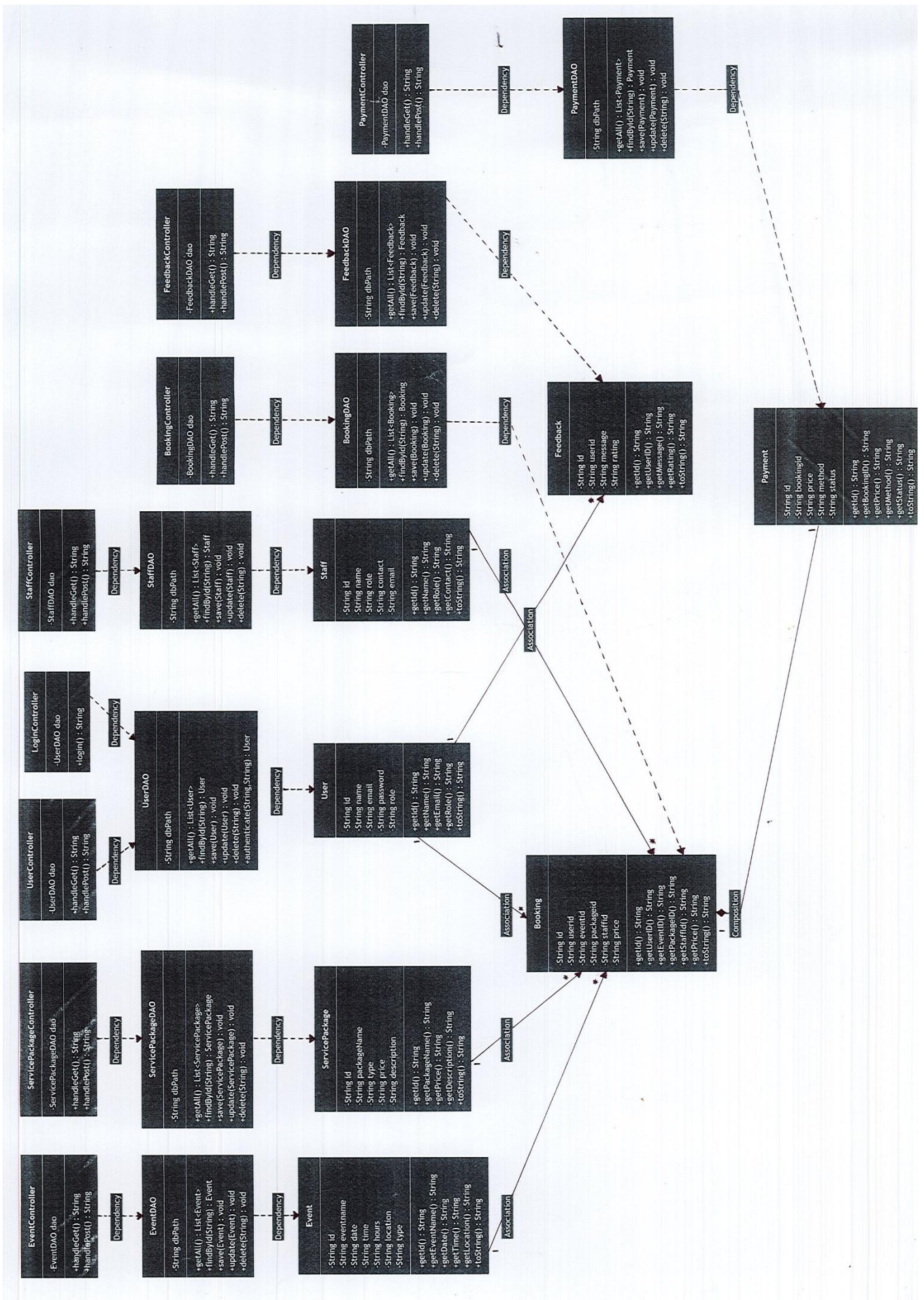
Update Operations

- Edit User Details
- Edit Events
- Update Packages
- Update Staff Details
- Reschedule Bookings
- Edit Feedbacks

Delete Operations

- Cancel Bookings
- Remove Users
- Delete Packages
- Delete Events

Class Diagram



Testing

The system was tested for:

- User Registration
- Login Authentication
- Event Booking Operations
- File Reading/Writing
- CRUD Operations

Testing Methods

- Manual Testing
- Functional Testing
- Input Validation Testing

GitHub Version Control

GitHub was used for:

- Source code management
- Team collaboration
- Commit tracking
- Backup management

GitHub Repository & Commit History

GitHub Repository : https://github.com/IT25100040/Event_Photography_and_Videography_Booking_System.git

Commit History

Team Contributions

Members	Contributions
IT25103466	Booking Managemen
IT25103641	Payment & Feedback Management
IT25102068	Staffs Management
IT25100040	Event Management
IT25102371	User Management
IT25100634	Package Management

Commits on May 14, 2026

init commit

Shankarban17 committed 2 days ago

4507b9

init commit

Shankarban17 committed 2 days ago

a5ac3d0

init commit

Shankarban17 committed 2 days ago

855d40f

init commit

Shankarban17 committed 2 days ago

3ba6f83

Commits on May 7, 2026

init commit

Shankarban17 committed last week

7e6d508

Commits on May 4, 2026

init commit

Shankarban17 committed 2 weeks ago

34c0ee9

Commits on May 1, 2026

init commit

Shankarban17 committed 2 weeks ago

c94dc59

Commits on Apr 29, 2026

init commit

Shankarban17 committed 2 weeks ago

881a22f

Commits on Apr 27, 2026

init commit

Shankarban17 committed 3 weeks ago

c97b2ac

Commits on Apr 26, 2026

init commit

Shankarban17 committed 3 weeks ago

8e3d5c7

init commit

Shankarban17 committed 3 weeks ago

6c2d484

Commits on Apr 25, 2026

init commit

Shankarban17 committed 3 weeks ago

f74d8ea

init commit

Shankarban17 committed 3 weeks ago

d59c36

init commit

Shankarban17 committed 3 weeks ago

c78c06

Commits on Apr 24, 2026

Initial commit

Shankarban17 committed 3 weeks ago

Verified

8d6c780

IT25103466

Commits on May 14, 2026

init commit

Shankarban17 committed 2 days ago

4507b9

init commit

Shankarban17 committed 2 days ago

a5ac3d0

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8d6c780

IT25103641

Advantages of the System

- Easy booking management
- Reduced paperwork
- Centralized records
- User-friendly interface
- Simple maintenance
- Lightweight storage
- Modular architecture

Limitations

- No SQL database integration
- Limited scalability for large businesses
- File storage may become slower with large datasets

Future Enhancements

Future improvements can include:

- MySQL or PostgreSQL integration
- Email notifications
- SMS alerts
- Mobile application development
- Cloud storage integration
- AI-based booking recommendations
- Real-time chat support

Conclusion

The Event Photography & Videography Booking System successfully demonstrates the practical implementation of Java-based web application development using Spring Boot and Object-Oriented Programming concepts. The project provides an efficient solution for managing photography bookings, customers, staff assignments, and payments within a centralized platform.

The system effectively automates traditional manual operations and improves business efficiency through digital management. The use of MVC architecture ensures clean code organization, while the file-based persistence mechanism demonstrates strong understanding of Java file handling concepts.

Overall, this project serves as an excellent educational and practical implementation of:

- Java Programming
- Spring Boot Development
- MVC Architecture
- File Handling
- CRUD Operations
- Object-Oriented Programming Principles

The project can be further expanded into a fully scalable commercial application with advanced features and database integration in the future.